CSE 4095 Team Project

Odin Stream

All your screens in one place.

What it is

A web app to discover movies and shows, track what you watch, and share ratings with friends.

Why we’re building it

We watch across many streaming services and lose track of episodes, watchlists, and where things are playing.

How it’s different

Film-first apps like Letterboxd focus on movies. Odin Stream tracks movies and shows, with episode progress and streaming availability, in one place.

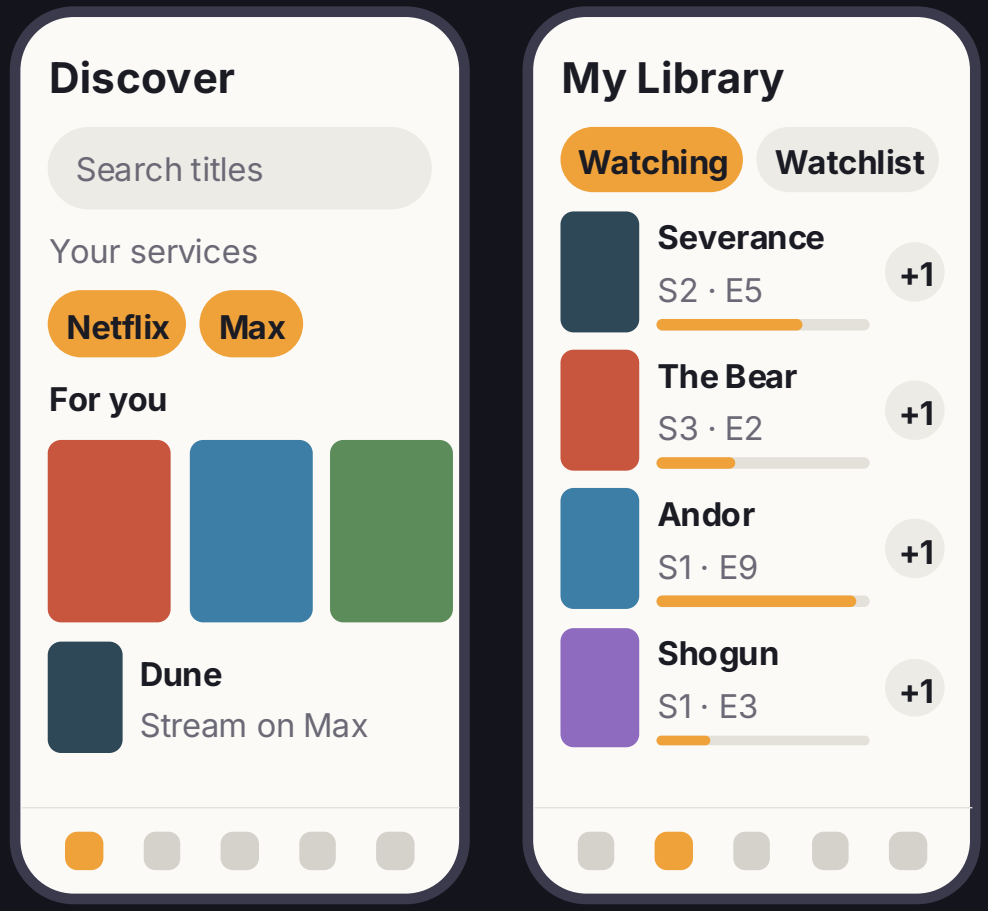
Discover and My Library

Discover

Search any movie or show, see where to stream, rent, or buy it in your country, and get recommendations that favor your services.

My Library

Keep a watched list and a watchlist. For shows, save your season and episode and pick up where you left off.



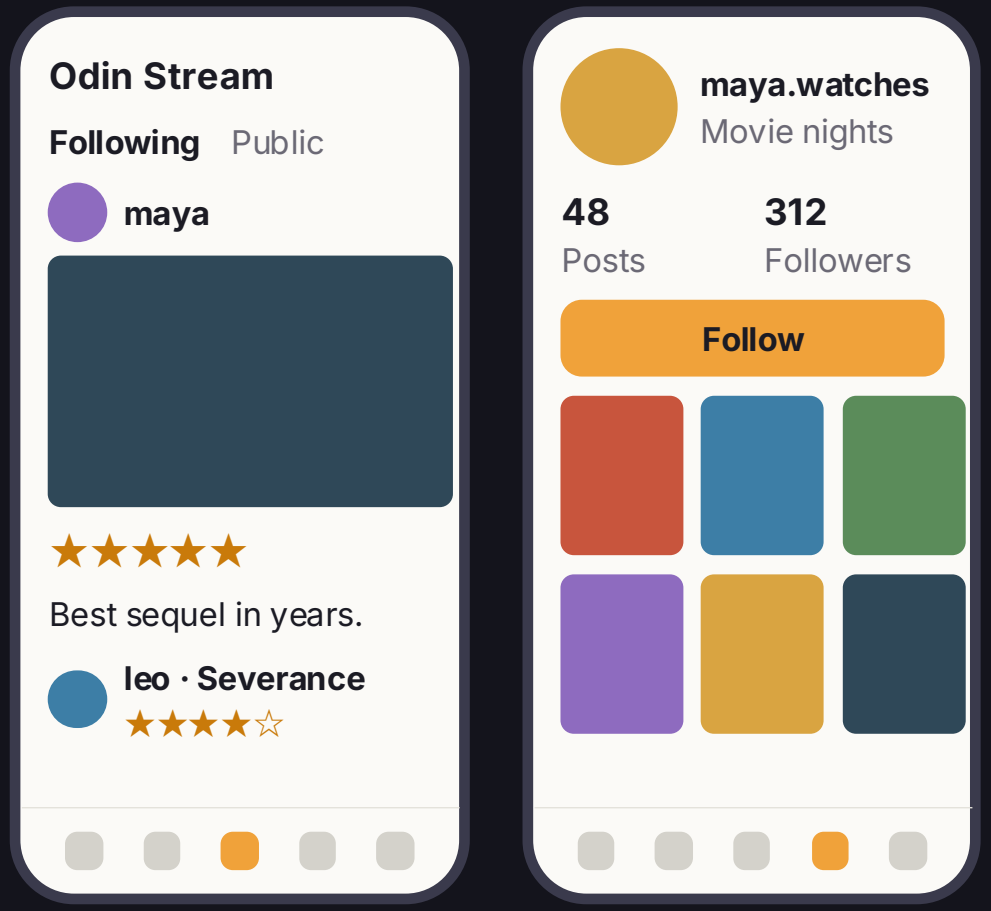
Feed and Profile

Feed

Follow other accounts and read their ratings and reviews, modeled on Instagram. Switch between Following and Public, and tap a post to open the title.

Profile

Every user has a public profile with their ratings and reviews. Follow or unfollow accounts, and set your country and streaming services to power recommendations.



Post and Recommendations

1 Pick a title

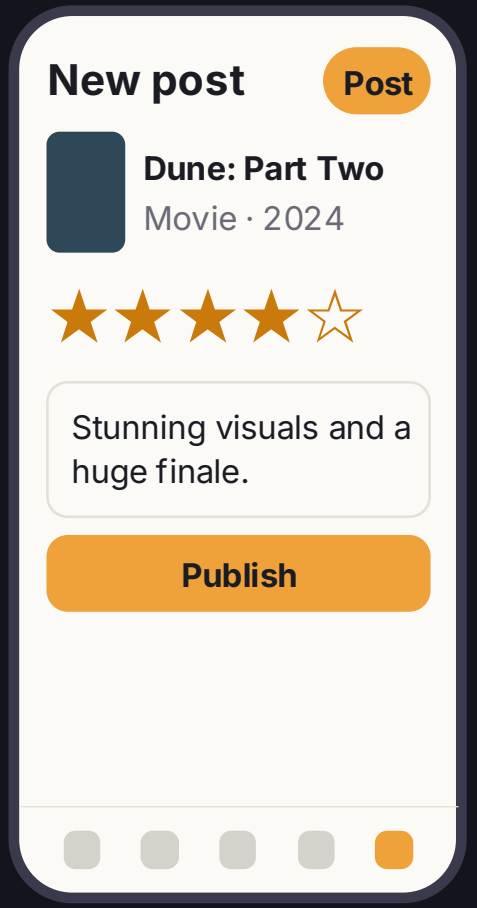
Choose a movie or show, rate it from one to five stars, and optionally write a review.

2 Share it

Posts appear on your profile, the public feed, and your followers' Following feed. You can edit or delete your own posts.

3 Improve recommendations

Saved ratings, your watchlist, and your streaming services help rank what to watch next.



Technology Stack

Front end

React, TypeScript, HTML, CSS

Back end

Python with FastAPI

Database

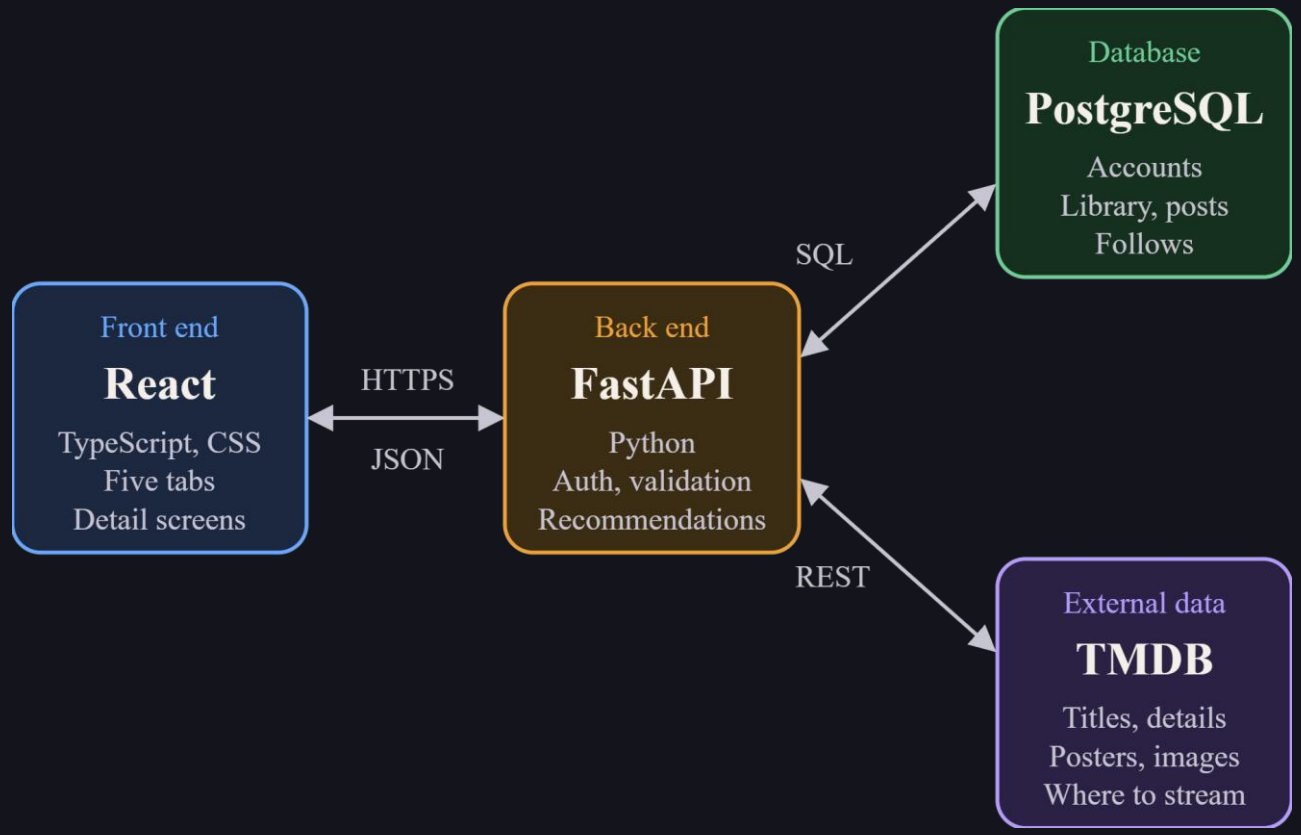
PostgreSQL

External data

TMDB, with JustWatch streaming availability

Dev tools

GitHub, with ChatGPT or Codex for help



How the Stack Works Together

Opening a title

1 · React
Search for Dune and open its page. React calls the API.

2 · FastAPI
Checks your login and validates the request.

3 · TMDB
Returns the poster, details, and where Dune streams in your country.

4 · PostgreSQL
FastAPI also reads your library, ratings, and services.

5 · React
Receives the JSON and draws the detail screen with Where to watch.

Posting a review

1 · React
Pick a title, choose a star rating, write a review, and tap Publish.

2 · FastAPI
Confirms who you are and validates the post.

3 · PostgreSQL
Saves the post, linked to the title's TMDB ID.

4 · FastAPI
Serves it to feeds and uses the rating to rank recommendations.

5 · React
Shows the post on your profile and in followers' feeds.

Development Timeline

Nov 4–10

Finalize scope, sketch the five tabs, set up the repo, verify API access

Nov 11–17

Accounts, search, title details, database storage

Nov 18–24

Library, watchlist, and episode progress

Nov 25–Dec 1

Profiles, following, posts, and both feeds

Dec 2–8

Recommendations, integration, testing, and demo rehearsal

Who builds what

Jack

Discover and My Library screens

Medha

Accounts, database, and TMDB integration

Raif

Feed, Profile, Post, and recommendations

Challenges and Feasibility

External data

Titles and streaming availability depend on TMDB and vary by country and over time.

Plan: Verify API access early, test several regions, and show a clear message when availability is missing.

Recommendation quality

New users have little history, and watch counts may not reflect taste.

Plan: Start with a simple score from genres, ratings, watchlist, and services; new users pick favorite genres.

Accounts and feeds

Progress and posts must belong to the right users and show up in the right feeds.

Plan: Validate requests on the server and test permissions, follows, post edits, and episode updates.

Feasibility

Achievable this semester: TMDB provides the catalog data, so we can focus on the interface, saved viewing data, social features, and recommendations. Optional extras (stats, reminders, spoiler controls, import and export) come after the five core features work together.